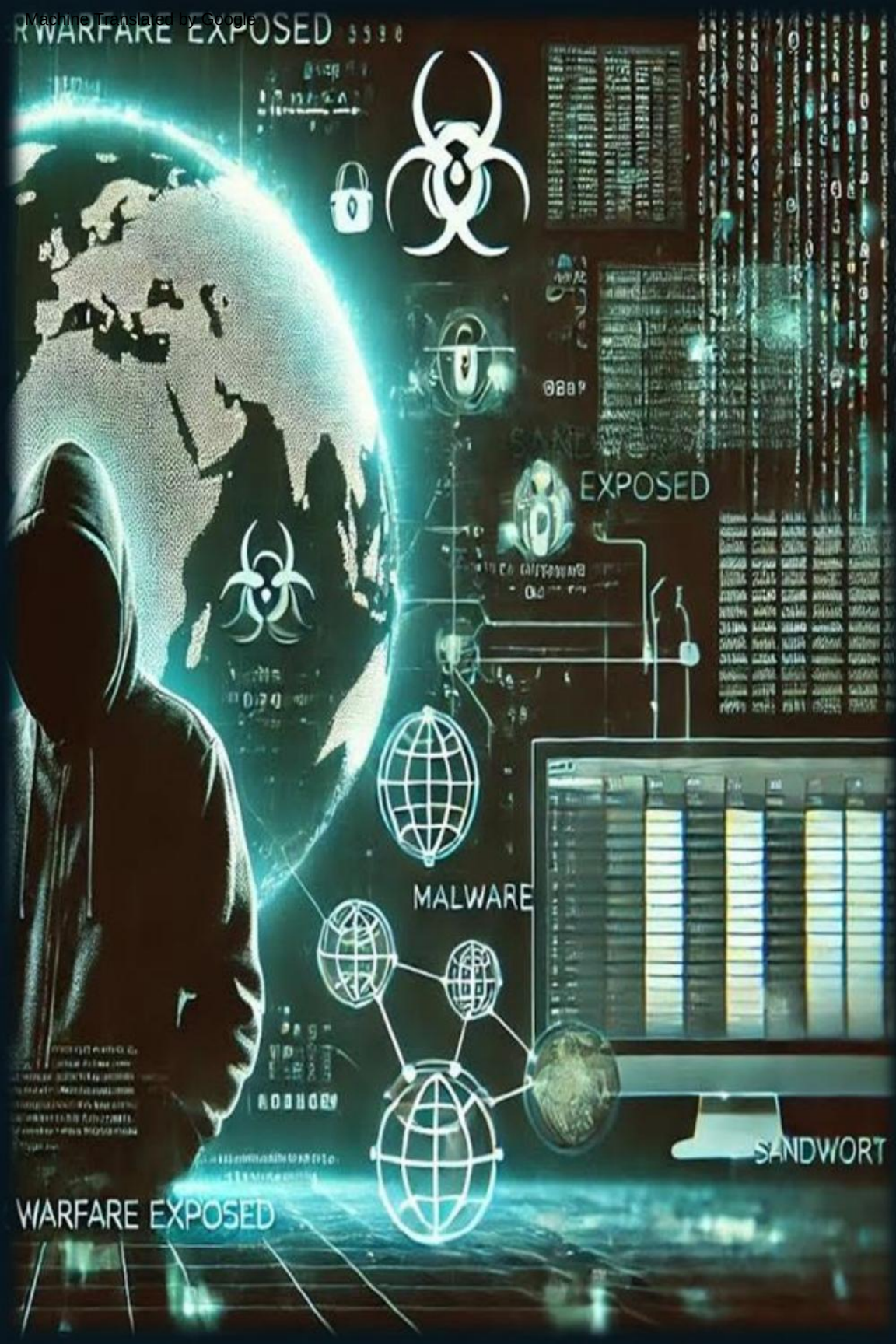




Final Project

Predicting Cyber Attacks at Microsoft

Presenters' names: Bar Cohen and Sahar Haim Yaakov

Submitted to: Mr. Avi Zakai

Moderator: Mr. Hanan Lev

From the Max Stern Jezreel Valley College

The project addresses the critical need for early prediction of cyber events targeting Microsoft systems. By leveraging data science techniques, we aim to develop a robust predictive model to improve cybersecurity posture and enable proactive decision-making for organizations.

The Growing Threat Landscape

The increasing frequency of cyberattacks requires advanced predictive capabilities. Our project addresses these challenges by focusing on Microsoft systems data, known as a common target for , malicious operations. The core goal is to predict cyber events using security data including 23,000 observations , thereby securing organizations' systems, minimizing potential damage, and optimizing their security strategies.



Security improvement

Improving protections against threats



Early prediction of incidents

Identify potential attack events before a point of impact

AI-Hur.



Resource management in the organization

Correct and efficient allocation of resources in the organization



Business goals and project objectives

In today’s digital landscape, information security is of paramount importance. This project aligns with business objectives to reduce response times and optimize critical resource allocation by improving the ability to predict cyberattacks . Success is measured not only by the accuracy of the prediction but also by the reduction of false positives and the overall reduction in the impact of cyber incidents.

+ 15%

Prediction accuracy

Accuracy target

5 minutes -

Early detection of attacks

Early detection and assessment of attacks

+ 25%

Resource optimization

Optimization and efficiency in resource allocation

Which translates directly into improved organizational resilience from before. These metrics ensure that our model delivers value to the organization's security, Cyber threats.



Methodology: DM-CRISP approach

Our project adopted the Data-Informed Standard Process (DM-CRISP) methodology, a structured approach that ensures comprehensive coverage from initial understanding to final deployment. This systematic framework guided us through every stage of the project, ensuring rigor and efficiency.

01

Business understanding

Clearly define the business problem and goals of the project.
Understanding the organizational context, the precise needs of decision makers
And the questions that need to be answered. Proper planning here saves time and resources in the following stages.

02

Understanding the data

Collecting primary data from various sources and conducting an initial investigation of their structure, quality, and content. The goal is to identify gaps, anomalies, errors, or patterns.
Interesting that can affect the surgery. Sometimes problems are discovered already at this stage

03

Data preparation

We will clean, adjust, and organize the data so that it is ready for modeling. This step includes handling missing values, converting data types, normalizing, filtering, and designing new variables.
This stage is often the longest in a data analysis project.

04

requiring additional data collection.

Modeling

We will select the appropriate modeling methods and adapt the model to the data.
And we will adjust parameters to improve performance.

05

Model evaluation

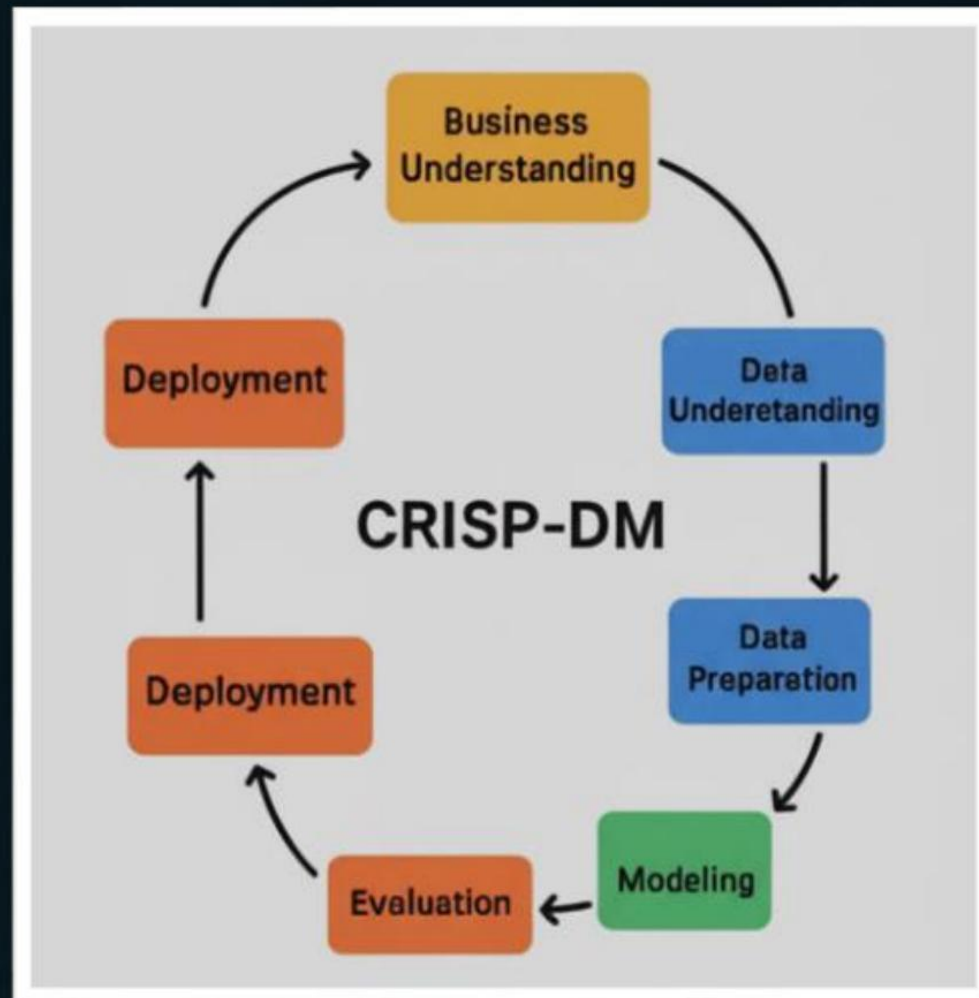
We will examine the quality of the model according to quantitative measures such as accuracy. In addition, we will check whether
Does the model actually meet the business goals defined at the beginning of the project? If not,
Return to the modeling or preparation phase for further improvements.

06

deployment

We will implement the model in the organization's work environment so that it can be used for real-time decision-making. The implementation. At this stage, we also plan for future maintenance and upgrades.

This approach provides a **clear and methodological framework** that helps data teams execute projects in an organized manner, go back if necessary, improve, and communicate the work process to clients or managers.



- Neat and circular frame
- Flexibility – returning to previous stages
 - Comprehensive business training
 - Comprehensive focus on data

Data focus and initial challenges

The foundation of our predictive model lies in comprehensive data. We obtained data primarily from official Microsoft security bulletins and related vulnerability databases. The collection involved manual downloads, and as a result of data integration, a dataset grew with dozens of features and tens of thousands of records.



- Initial testing revealed common data quality issues

- Missing, duplicate, inconsistent or typographical data.

"Severity.1" Severity"

- Duplicate or overlapping columns (e.g.,

- Incorrect data and dates

- These challenges were systematically addressed through rigorous data cleaning and validation processes to ensure data integrity and reliability for subsequent analysis.

Data preparation

After the initial cleaning, the dataset underwent extensive preparation and comprehensive data analysis. Statistical analysis helped us understand the distributions of variables, such as threat types and severity levels. We used various tools to highlight trends and patterns, especially how threat severity evolves over time.

Feature selection

Identifying and selecting the most relevant features for training models according to p-value and statistical measures.

Handling values

Handling missing values by filling in the most common value and predicting missing target values using the Random Forest model and date separation.

Data transformation

To improve , Normalization, categorical value encoding, and new feature engineering increase model efficiency and improve the accuracy of the selected model.

Data integration

Enriching the data set by combining information from 2 Excel files according to existing attributes and identical columns.

Using graphs, tables, and more, show vital insights

about common attack types and their relationship to severity and frequency, and we made assumptions and decisions about the types of models we used.



```
def fill_missing_categorical_values(df, how='most_frequent'):
#Fills missing values in a DataFrame's categorical columns
# using the specified imputation strategy
df = df.replace({None: np.nan})
imputer = SimpleImputer(strategy=how,
                        fill_value='missing')
filled_array = imputer.fit_transform(df)
df = pd.DataFrame(filled_array, columns=df.columns)
return df
```

Data cleaning - handling missing values

The data cleaning step and handling missing values of features and the target variable is a critical step in the data processing process, because it ensures that the model receives high-quality, consistent input without biases resulting from missing information. This step may directly affect the accuracy, reliability, and generalizability of the model in machine learning.

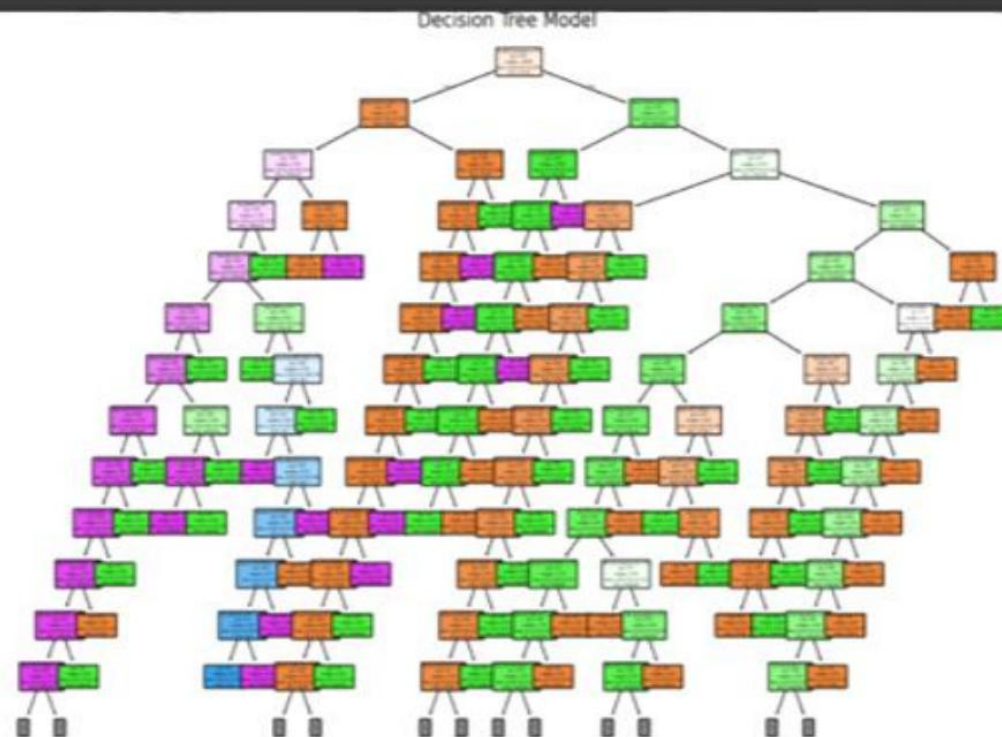
Separating the date column.

- Filling missing values in model attribute columns (without the target feature) with the column's common value.

- Combining the target column and separating into a group of missing and non-missing values (group

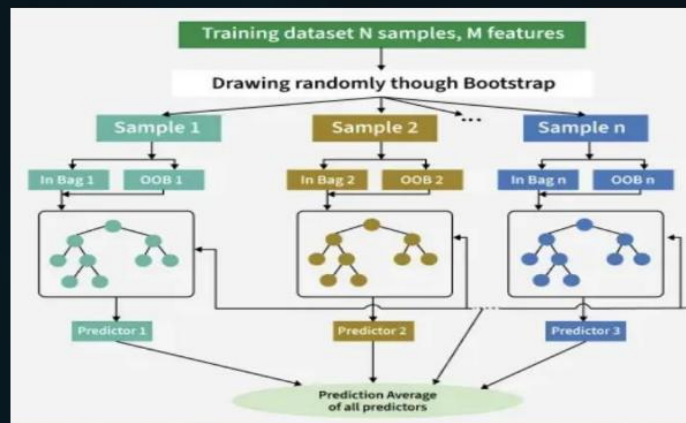
Experiment and prediction.

- **Prediction and prediction of the prediction group** (which is divided into training and testing) by the experimental group using a ready-made learning model.



Models

We evaluated several advanced machine learning models to identify the best fit for our prediction task. The basic operation of each model was examined and their performance was accurately evaluated and tested. The dataset was divided into training, validation, and testing sets to ensure robust evaluation and avoid overfitting.

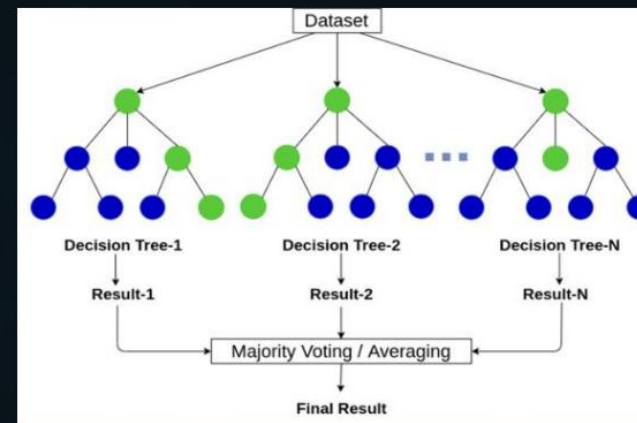


CatBoost

Improving the decision tree based on previous trees

and error reduction.

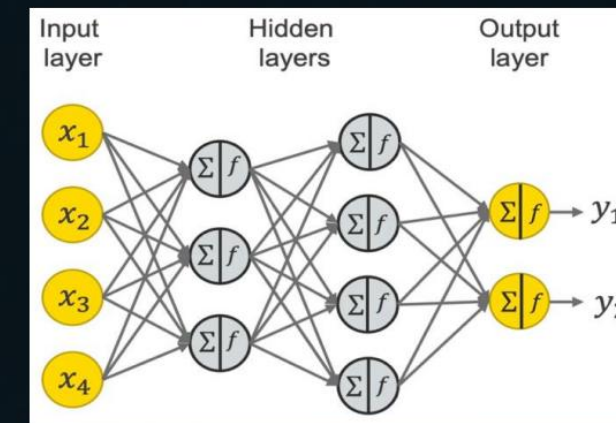
Automatic handling of missing values



Random Forest

Combining multiple decision trees for accuracy

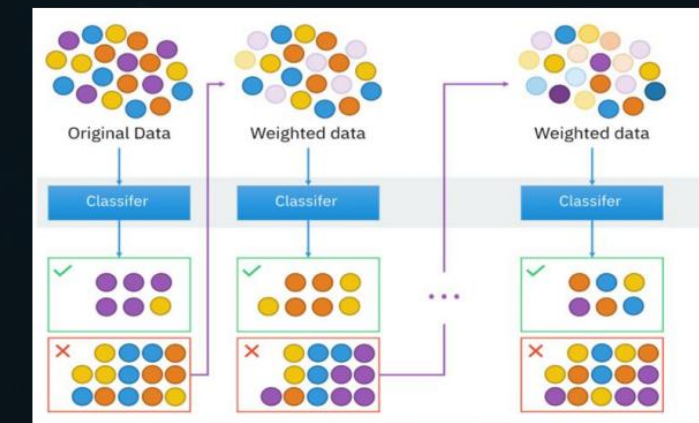
improved.



Neural Networks

Deep learning model for pattern recognition

Complex in data



AdaBoost

Using weak models to create a model

Strengthening and improving the model

Parameter tuning, conducted using cutting-edge tools such as Optuna (automatic tuning of model parameters), was essential to optimizing the performance of each model, ensuring that they would realize the model's potential in predicting cyber events.



Model evaluation

Initial results showed varying performance across

The models that were evaluated, some of which significantly outperformed

Others. A comprehensive comparison of key metrics provided an understanding

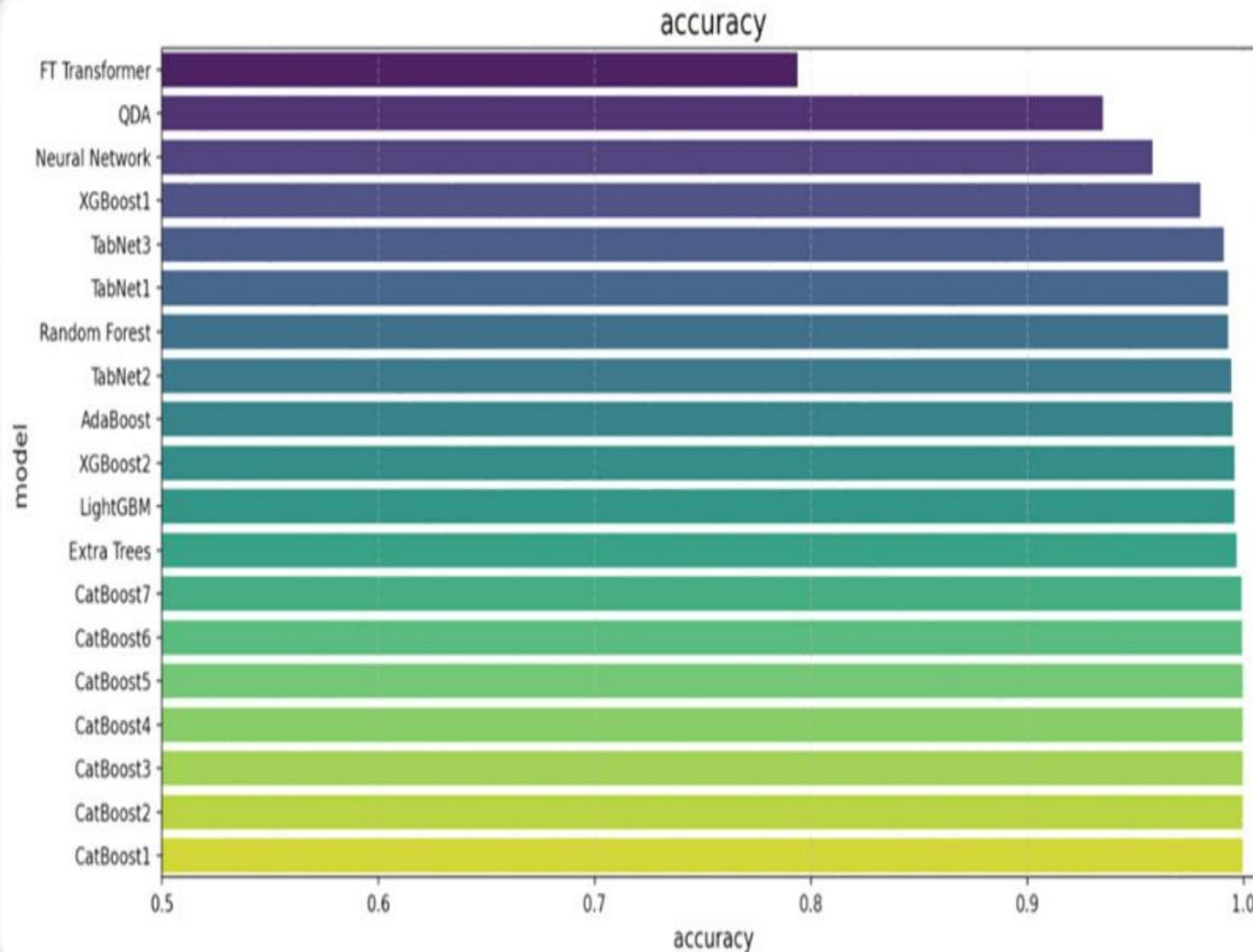
A clear understanding of the strengths and weaknesses of each model.

Confusion matrices and performance curves were essential in simulating

The results, in addition to understanding the characteristics that influence

The model.

Using Optuna, run different models with different parameters to find the best
model.



Selected model - CatBoost

- The model handles categorical features

Automatically, without the need for coding

manual.

- Building a tree to reduce the error of
The previous model, by number of runs.

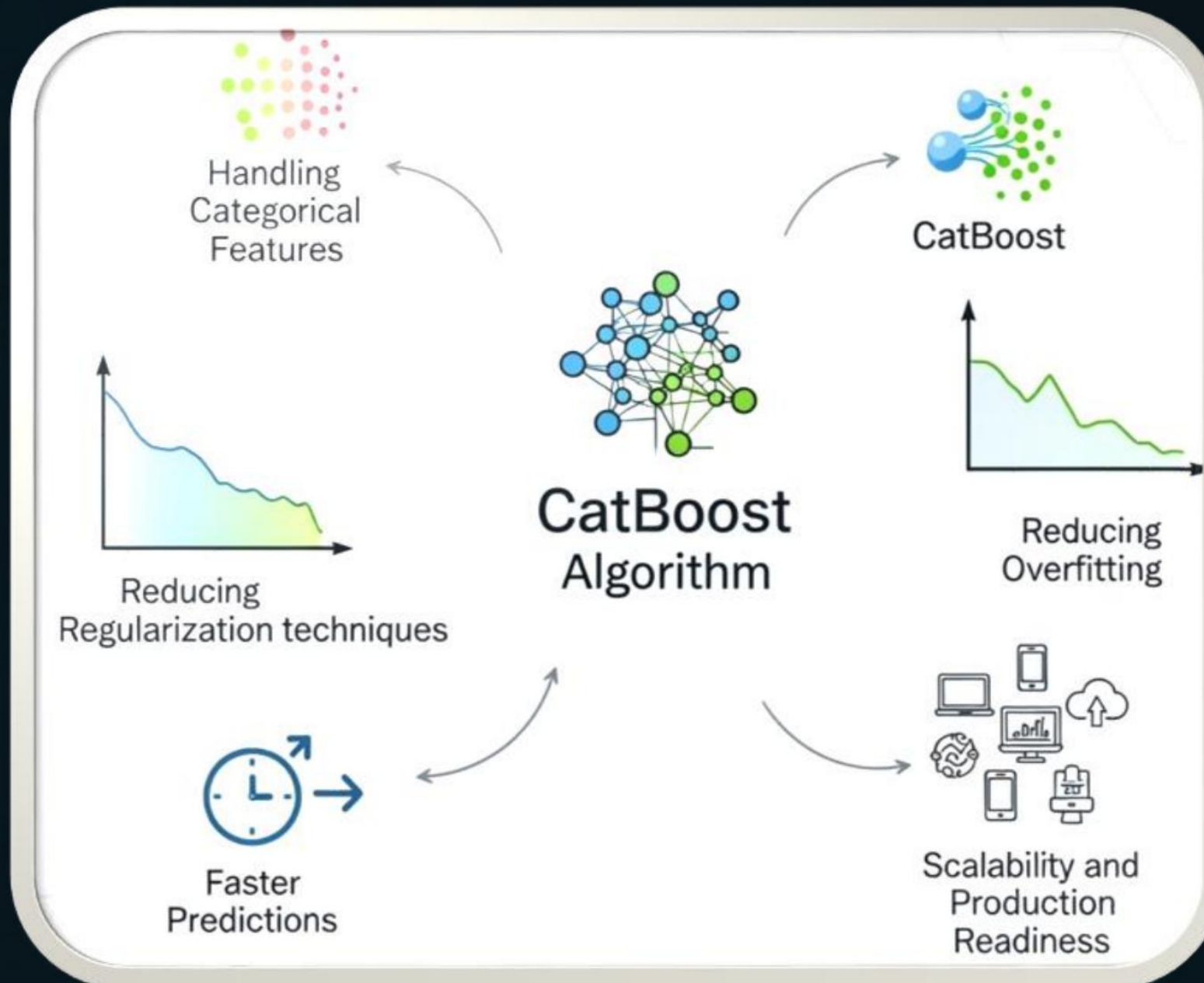
- Uses the technique he focuses on

In correcting errors in previous trees by

.loss function

- Looking for the best wood by using

In gradient.



Deployment and maintenance plan

The final deployment plan ensures the accessibility and scalability of our predictive model. The system architecture is designed to support continuous operation and user interaction, making the model a practical tool for cybersecurity professionals.

Code Hosting



All project code is hosted on GitHub for version control and collaboration.

Dashboards



Interactive dashboards for data visualization in Tableau

Data Storage



The data is stored securely in a MYSQL database.

Interactive Interface



Interaction with the user is made possible through platforms such as .Gradio , Flask , Hugging Face

For continued success, a robust maintenance plan is in place, including continuous monitoring of data quality, model accuracy, and system performance. This plan also outlines methods for updating the model, retraining with new data, and identifying anomalies or major changes in the environment to ensure the model remains effective and relevant.



Summary

The project developed a predictive model for Microsoft-related cybersecurity events that demonstrates measurable business value and lays the foundation for future improvements. The entire process provided valuable insights into data-driven security analytics and the complexities of deploying machine learning in the real world.

"The future of cybersecurity lies in proactive prediction, not just reactive defense."



We are confident that this model will make a significant contribution to strengthening cybersecurity postures and enabling more informed and proactive decisions for organizations. We welcome any questions or feedback as we continue to refine and expand this critical work.



Links



[GitHub](#)



Hugging Face - [Gradio](#)

[Flask](#)



[Collaborate](#)



Dashboard – [1](#) , [2](#)



Back to the beginning

The presentation



[Bar Cohen](#)



[Sahar Haim Yaccov](#)



final project